Transaction Tracker - Oracle Database Design

Overview

This document outlines the Oracle database design for the Transaction Tracker application. The design focuses on scalability, normalization, and multi-user support.

Core Tables

1. USERS

```
USERS (
    USER_ID         NUMBER PRIMARY KEY,
    EMAIL           VARCHAR2(255) UNIQUE NOT NULL,
    NAME            VARCHAR2(255),
    CREATED_AT      TIMESTAMP DEFAULT CURRENT_TIMESTAMP
);
```

2. TRANSACTIONS

```
TRANSACTIONS (
    TRANSACTION_ID   NUMBER PRIMARY KEY,
    USER_ID          NUMBER NOT NULL,
    AMOUNT           NUMBER(12,2) NOT NULL,
    TRANSACTION_DATE DATE,
    DESCRIPTION      VARCHAR2(500),
    CATEGORY_ID      NUMBER,
    CREATED_AT       TIMESTAMP DEFAULT CURRENT_TIMESTAMP,

    CONSTRAINT fk_user FOREIGN KEY (USER_ID) REFERENCES USERS(USER_ID),
    CONSTRAINT fk_category FOREIGN KEY (CATEGORY_ID) REFERENCES CATEGORIES(CATEGORY_ID)
);
```

3. CATEGORIES

```
CATEGORIES (
    CATEGORY_ID     NUMBER PRIMARY KEY,
    USER_ID         NUMBER,
    NAME            VARCHAR2(100) NOT NULL,
    PARENT_ID       NUMBER,

    CONSTRAINT fk_parent_category FOREIGN KEY (PARENT_ID) REFERENCES CATEGORIES(CATEGORY_ID)
);
```

4. UPLOADS

```
UPLOADS (
    UPLOAD_ID      NUMBER PRIMARY KEY,
    USER_ID        NUMBER,
```

```
    FILE_NAME     VARCHAR2(255),
    UPLOADED_AT   TIMESTAMP DEFAULT CURRENT_TIMESTAMP
);
```

5. TRANSACTION_UPLOAD_MAP

```
TRANSACTION_UPLOAD_MAP (
    TRANSACTION_ID NUMBER,
    UPLOAD_ID      NUMBER,

    PRIMARY KEY (TRANSACTION_ID, UPLOAD_ID),
    FOREIGN KEY (TRANSACTION_ID) REFERENCES TRANSACTIONS(TRANSACTION_ID),
    FOREIGN KEY (UPLOAD_ID) REFERENCES UPLOADS(UPLOAD_ID)
);
```

Indexing Strategy

```
CREATE INDEX idx_transactions_user ON TRANSACTIONS(USER_ID);
CREATE INDEX idx_transactions_date ON TRANSACTIONS(TRANSACTION_DATE);
CREATE INDEX idx_category_user ON CATEGORIES(USER_ID);
```

Future Enhancements

- Partition TRANSACTIONS table by date - Add tagging support - ML-based categorization - Data encryption per user